

Status	Finished
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Completed	Friday, 31 July 2026, 1:59 PM
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Information

Examiners: K. Matekaire

Moderators: B. Mpofu

Time: 90 minutes

Marks: 85 Marks

Instructions

No instructions or directives of the examiner shall be disregarded; if any of the instructions are disobeyed, candidates shall expose themselves to disqualification from future examinations.

General

- All questions must be answered on the specified platform.
- Use the **Question Navigation** blocks on the right-side of the page to navigate through questions.
- Results of the questions appear soon after the submission has been made.
- Plagiarism is a serious offense and can result in disciplinary action.
- You are **not allowed** to wear headphones, earphones, hats, or hoodies.
- You may not communicate with anyone in the environment you are in.
- You are not allowed to use an additional screen and should keep your attention on the screen in front of you.
- Candidates shall not communicate or attempt to communicate with anyone and candidates shall not conduct themselves in an improper or unseemly manner.
- Each candidate is expected to maintain academic integrity. Any form of plagiarism or cheating that is detected can lead to disciplinary measures being taken.

Technical

- Any technical issue related to the examination platform being used needs to be reported to the invigilator immediately.

Question 1

Complete

Marked out of 1.00

What is the primary purpose of a Dockerfile?

- ☐ a. To store Docker container logs.
- ☐ b. To manage Docker volumes.
- ☐ c. To orchestrate multiple Docker containers.
- ☒ d. To define how a Docker image is built.

Question 2

Complete

Marked out of 1.00

What does the RUN instruction in a Dockerfile typically execute?

- ☐ a. Commands to copy files into the image.
- ☐ b. Commands during the image build process.
- ☐ c. Commands to expose ports.
- ☒ d. A command when the container starts.

Question 3

Complete

Marked out of 1.00

What is the primary function of a Docker image?

- ☐ a. To manage running containers.
- ☒ b. To be a lightweight, standalone, executable package of software that includes everything needed to run an application.
- ☐ c. To store application data persistently.
- ☐ d. To create an instance of a running application.

Question 4

Complete

Marked out of 1.00

What is a Docker container?

- ☐ a. A template for creating images.
- ☐ b. A registry for Docker images.
- ☒ c. A running instance of a Docker image
- ☐ d. A tool for building Dockerfiles.

Question 5

Complete

Marked out of 1.00

How does Docker achieve isolation for containers?

- ☐ a. By creating a full virtual machine for each container.
- ☐ b. By leveraging kernel features like namespaces and control groups.
- ☒ c. By running a separate operating system within each container.
- ☐ d. By requiring separate hardware for each container.

Question 6

Complete

Marked out of 1.00

Which instruction in a Dockerfile is used to specify the base image for the build process?

- ☐ a. RUN
- ☒ b. FROM
- ☐ c. CMD
- ☐ d. EXPOSE

Question 7

Complete

Marked out of 1.00

What is the primary purpose of the `ENTRYPOINT` instruction in a Dockerfile for a Java application?

- ☒ a. To specify the base image from which to build.
- ☐ b. To compile the Java source code into bytecode.
- ☐ c. To define the main command to be executed when a container is started.
- ☐ d. To copy the Java source code into the Docker image.

Question 8

Complete

Marked out of 1.00

When running a Docker container, if the required image is not available locally, what is the *immediate* next step Docker takes after checking local storage?

- ☐ a. Executes the specified command in an isolated container.
- ☒ b. Pulls the image from the registry.
- ☐ c. Checks for an updated version of the image.
- ☐ d. Prompts the user to build the image.

Question 9

Complete

Marked out of 1.00

A Dockerfile includes the line `FROM mcr.microsoft.com/windows/servercore:ltsc2019`. What is the primary purpose of this

- ☒ a. To set the base image for the Docker image.
- ☐ b. To install the IIS web server role.
- ☐ c. To create an HTML file within the image.
- ☐ d. To specify the maintainer of the image.

Question 10

Complete

Marked out of 1.00

Which command is used to download a Docker image from a registry?

- ☐ a. docker build
- ☒ b. docker pull
- ☐ c. docker push
- ☐ d. docker run

Question 11

Complete

Marked out of 1.00

The EXPOSE instruction actually publishes the port to the host system.

- ☒ True
- ☐ False

Question 12

Complete

Marked out of 1.00

The **docker build** command requires the path to the Dockerfile.

- ☒ True
- ☐ False

Question 13

Complete

Marked out of 1.00

A Docker image is a running instance of a Docker container.

- ☐ True
- ☒ False

Question 14

Complete

Marked out of 1.00

The COPY instruction is used to copy files *from* the Docker image *to* the host system

- ☐ True
- ☒ False

Question 15

Complete

Marked out of 4.00

Select **all correct options** for each Docker concept listed below. Multiple answers **may be correct** for each row.

	Runs on Host Kernel	Requires Full OS	Uses Layered Filesystem	Lightweight	Isolated Process Space
Virtual Machine (VM)	☒	☒	☐	☐	☐
Docker Container	☐	☐	☒	☒	☒
Docker Image	☒	☐	☒	☒	☒

Question 16

Complete

Marked out of 5.00

Match each term in Column A with its best description or function in Column B.

Specifies the base image for the build process.

FROM

Executes commands during the image build process.

RUN

Provides defaults for an executing container.

CMD

Declares that the container listens on specified network ports at runtime.

EXPOSE

Copies new files or directories from a source path to a destination path inside the image.

COPY

Question 17

Complete

Marked out of 5.00

Multiple Response Question (Select ALL that apply)

Which of the following are considered best practices when writing a Dockerfile?

- ☐ a. Tightly coupled services in multiple containers
- ☒ b. Pin base image versions
- ☒ c. Use `.dockerignore` to exclude files
- ☐ d. Always install a text editor for maintenance
- ☐ e. Combine web app, DB, and cache in one container
- ☐ f. Install all possible libraries in advance
- ☒ g. Decouple services into separate containers
- ☐ h. Overwrite ENTRYPOINT during container startup
- ☒ i. Use multi-stage builds
- ☒ j. Sort multi-line arguments alphabetically

Question 18

Complete

Marked out of 5.00

Select ALL the options for the `docker container ls` command that can be used to filter or modify the output:

- ☐ a. `--volumes`
- ☒ b. `--no-trunc`
- ☒ c. `--last`
- ☐ d. `--quiet`
- ☐ e. `--size`
- ☒ f. `--filter`
- ☐ g. `--force`
- ☒ h. `--latest`
- ☐ i. `--digests`
- ☒ j. `--all`

Question 19

Complete

Marked out of 13.00

Fill in the missing Docker CLI commands in the gaps.**Use shorthand commands where necessary; dashes are included for you.**

1. The command to build a Docker image with the tag myapp is: **docker** -t myapp.
2. To list all running containers, use: **docker**
3. To stop a running container by ID or name: **docker** <container_id>
4. The command to start a container in detached mode is: docker - <image>
5. The command to remove a container is: **docker** <container_id>
6. To display logs from a running container: **docker** <container_id>
7. To see all containers, including stopped ones: **docker ps** -
8. To remove an image: **docker** <image_id>
9. To remove all stopped containers: **docker** -f
10. To inspect container configuration details: **docker** <container_id>
11. The command to login to Docker Hub from CLI: **docker**

Question 20

Complete

Marked out of 4.00

Fill in the blanks to complete the Dockerfile by dragging and dropping the correct keywords into place.

python:3.10-slim
 /app
COPY . .
CMD [" ", " "]

Question 21

Complete

Marked out of 3.00

You are tasked with automating the deployment of a Python-based microservice using a **Python script** and a **text file** that contains a list of Docker image names. Each line in the file corresponds to a container that must be pulled, started, and monitored individually. Put the following steps in the correct order to complete the task using a **Python script** and **Docker CLI**, assuming the text file is named `images.txt`:

- a. Use Python's `open()` function to read each line from `images.txt` into a list
- b. Strip newlines and validate that each image name is not empty before running Docker commands
- c. Use `subprocess.run()` to execute `docker pull` for each image
- d. Start each container using `docker run -d` and capture container IDs using `subprocess`
- e. Check the status of each container using `docker inspect` or `docker ps` to confirm they are running
- f. Save container IDs to a log file called `running_containers.log` for future reference

Question 22

Complete

Marked out of 15.00

For each question, read the scenario carefully and then write the complete Docker command that would achieve the described outcome in the space provided. Pay attention to all options and flags mentioned in the scenario.

N.B Please type exactly how the command appears.

1. You've just pulled a new image named my-api. What's the basic command to run a container from this image?	docker	run	
		my-api	
2. How do you list all Docker containers, including those that are stopped or exited?	docker	ps	-a
3. You have a container named web-server that is currently running. What command stops it gracefully, allowing time for cleanup before forcing termination if necessary?	docker	stop	
		web-server	
4. To remove a specific Docker image named old-app-image (assuming no containers are using it), what command would you use?	docker	rmi	old-app-image
5. You need to delete a stopped container named temp-data-processor permanently. What command achieves this?	docker	rm	temp-data-processor
6. To display detailed, low-level information about a specific Docker image named alpine:3.14, including its configuration, layers, and metadata, what command would you use?	docker	inspect	
		alpine:3.14	
7. After building a local image, you want to push your image yourusername/my-microservice:v1.0 to Docker Hub. Which command is used for this purpose?	docker	push	
		yourusername/my-microservice:v1.0	
8. You want to reclaim the maximum amount of disk space by removing all stopped containers, dangling images, unused networks, and build cache. Which command should you use?	docker	system	
		prune	
9. You specifically want to delete only stopped Docker containers to free up space. Write the appropriate command.	docker	container	
		prune	
10. You have a Docker image tagged my-app:dev that you want to rename to my-app:latest without creating a new image. Write the appropriate command	docker	-t	my-app:dev my-app:latest

Question 23

Complete

Marked out of 17.00

Important Note:

- Only include instructions and code in the Dockerfile that are **necessary to achieve the required functionality**.
- Marks will be **deducted** for any unnecessary or irrelevant code, commands, or comments that do not contribute to building and running the container correctly.
- Focus on clarity, correctness, and adherence to the instructions provided.
- Complete all the missing parts in the Dockerfile below by writing the correct code directly into the spaces provided.

1. Given the following simple Python script called hello.py that prints "Hello, World!".

Python script (testapp.py)

```
# testapp.py
print("Hello from testapp.py!")
```

2. Complete the Dockerfile according to the following instructions: Each gap has 1 mark awarded

1. Use the official Python 3.10 image.
2. Set the author and maintainer labels as **"Thomas Cook"** and email **"Cook.T@belgiumcampus.ac.za"**.
3. Copy **testapp.py** into the root directory of the Docker image so it is available at **/testapp.py**.
4. Ensure that when the container starts, it automatically executes the Python script using the correct interpreter.
5. Add an instruction to expose port **8080**.
6. Provide a brief comment in the Dockerfile on why exposing ports can be useful for Docker containers.
7. Set an environment variable named **APP_ENV** with the value **production**.
8. Add a health check that verifies the container responds on port 8080 using **curl**.
9. Create a non-root user named **appuser** and switch to that user for security purposes.
10. Add a build-time argument named **VERSION** and set it as an environment variable inside the container.

1. Use the official Python 3.10 image

FROM

python 3.10

2. Set author and maintainer labels

LABEL AUTHOR="Thomas Cook" \

email="Cook.T@belgiumcampus.ac.za"

3. Copy testapp.py into the root directory

COPY testapp.py .

4. Expose port 8080

EXPOSE -p

8080

Comment: Exposing ports is usefull to ensure the docker file/image knows where it is allowed to

5. Set environment variable

ENV

APP_ENV= production

6. Add health check

CMD curl --fail http://localhost:8080 || exit 1

7. Create non-root user and switch

CREATE

useradd -m appuser

CREATE

appuser

8. Add build-time argument and use it

ADD

VERSION=1.0

ADD

VERSION=\$VERSION

9. Set default command to run the Python script

CMD

["python" , "/testapp.py"]

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